

AMAR TELIDJI UNIVERSITY

Laghouat

Computer Sciences Department

Class: **4th Year Cybersecurity**

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COURSE

Cyber Security

REPORT TITLE

Lab Report

2025 / 2026

Introduction

This lab focuses on applying practical cybersecurity concepts through different defensive and investigative tools. The exercises cover several important areas, including network scanning, encryption, steganography, password security, breach checking, onion service analysis, firewall hardening, IDS/IPS configuration, antivirus integration, log collection, incident ticketing, and security automation. The goal is to understand how these tools are used in real environments to identify risks, protect systems, monitor activity, and respond to security incidents.

Objectives

The main objectives of this lab are to:

- identify exposed services and possible vulnerabilities using network scanning tools;
- protect sensitive files using encryption and data concealment techniques;
- improve password security using vaults, password policies, aging rules, and breach checks;
- analyze hidden services and understand metadata exposure risks;
- configure OPNsense as a layered security firewall using IDS/IPS, NGFW, and antivirus tools;
- collect, centralize, and visualize system logs using the ELK Stack;
- detect anomalies such as failed login attempts and privilege-related events through log analysis;
- create and manage incident cases using a ticketing system;
- understand how cybersecurity tools support detection, prevention, accountability, and traceability.

Exercise 01:

```
(kali㉿kali)-[~]
$ nmap -sn 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 11:52 CET
Nmap scan report for 192.168.1.10
Host is up (0.0018s latency).
MAC Address: 08:00:27:B5:BF:D2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds

(kali㉿kali)-[~]
$ nmap -sV 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 11:52 CET
Nmap scan report for 192.168.1.10
Host is up (0.0012s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.1 (Ubuntu Linux; protocol 2.0)
23/tcp    open  telnet   Linux telnetd
9200/tcp  open  ssl/http Elasticsearch REST API 7.0 or later (Shield plugin; realm: security)
MAC Address: 08:00:27:B5:BF:D2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.62 seconds
```

- **nmap -sn** showing **Host is up**
- **nmap -sV** showing the open ports and services

```
(kali㉿kali)-[~]
$ nmap -sS -sV -O --script vuln -oN nmap_report.txt 192.168.1.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 11:53 CET
Stats: 0:08:28 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.43% done; ETC: 12:02 (0:00:03 remaining)
Stats: 0:08:28 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.43% done; ETC: 12:02 (0:00:03 remaining)
Nmap scan report for 192.168.1.10
Host is up (0.0019s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
| vulners:
|   vsftpd 3.0.3:
|   | CVE-2021-30047 7.5 https://vulners.com/cve/CVE-2021-30047
|   | CVE-2021-3618 7.4 https://vulners.com/cve/CVE-2021-3618
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.1 (Ubuntu Linux; protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:8.2p1:
|   | PACKETSTORM:173661 9.8 https://vulners.com/packetstorm/PACKETSTORM:173661 *EXPLOIT*
|   | F0979183-AE88-53B4-86CF-3AF0523F3807 9.8 https://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0523F3807 *EXPLOIT*
|   | CVE-2023-38408 9.8 https://vulners.com/cve/CVE-2023-38408
|   | B8190CDB-3EB9-5631-9828-8064A1575B23 9.8 https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575B23 *EXPLOIT*
|   | 8FC9C5AB-3968-5F3C-825E-E8DB5379A623 9.8 https://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB5379A623 *EXPLOIT*
|   | 8AD01159-548E-546E-AA87-2DE89F3927EC 9.8 https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927EC *EXPLOIT*
|   | 6192C35D-F78B-5C0A-AB8D-9826A79A5320 9.8 https://vulners.com/githubexploit/6192C35D-F78B-5C0A-AB8D-9826A79A5320 *EXPLOIT*
|   | 2227729D-6700-5C8F-8930-1EEAFD4B9FF0 9.8 https://vulners.com/githubexploit/2227729D-6700-5C8F-8930-1EEAFD4B9FF0 *EXPLOIT*
|   | 0221525F-07F5-5790-912D-F489E2D1B587 9.8 https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F489E2D1B587 *EXPLOIT*
|   | CVE-2026-35414 8.1 https://vulners.com/cve/CVE-2026-35414
|   | CVE-2026-35386 8.1 https://vulners.com/cve/CVE-2026-35386
|   | CVE-2026-35385 8.1 https://vulners.com/cve/CVE-2026-35385
|   | BA38878D-F579-53B1-A4A4-FF49E953E1C0 8.1 https://vulners.com/githubexploit/BA38878D-F579-53B1-A4A4-FF49E953E1C0 *EXPLOIT*
|   | 4FB01800-F993-5CAF-BD57-D7E290D10C1F 8.1 https://vulners.com/githubexploit/4FB01800-F993-5CAF-BD57-D7E290D10C1F *EXPLOIT*
|   | CVE-2020-15778 7.8 https://vulners.com/cve/CVE-2020-15778
|   | C94132FD-1FA5-5342-B6EE-0DAF45EEFFE3 7.8 https://vulners.com/githubexploit/C94132FD-1FA5-5342-B6EE-0DAF45EEFFE3 *EXPLOIT*
|   | 991D2CCA-0E09-5745-97A2-4917461BD6EC 7.8 https://vulners.com/githubexploit/991D2CCA-0E09-5745-97A2-4917461BD6EC *EXPLOIT*
|   | 2E719186-2FED-58A8-A150-762EFBAA523 7.8 https://vulners.com/gitee/2E719186-2FED-58A8-A150-762EFBAA523 *EXPLOIT*
```

- **nmap -sS -sV -O --script vuln -oN nmap_report.txt**
showing the detected vulnerabilities and OS guess

Extracted Information

Nmap was used to scan the Ubuntu target at [192.168.1.10](#). The scan confirmed that the host is reachable and identified four open ports:

[21](#) for FTP ([vsftpd 3.0.3](#)), [22](#) for SSH ([OpenSSH 8.2p1](#)), [23](#) for Telnet, and [9200](#) for Elasticsearch REST API.

Analysis and Vulnerability

These results show that several services are exposed on the target machine, which increases the attack surface. The vulnerability scan reported possible CVEs related to the FTP and SSH services, and Telnet is especially important because it is an insecure protocol that should not be exposed.

Exploitation

An attacker can use this scan information to identify the running services, search for known exploits, and attempt unauthorized access. Exposed SSH may be targeted with brute-force or known vulnerabilities, Telnet may be abused because it is insecure, and Elasticsearch may be queried if access controls are weak.

Protection

To reduce the risk, unnecessary services should be disabled, especially Telnet, exposed ports should be restricted with a firewall, and all services should be updated to secure versions. Sensitive services should only be accessible from trusted systems.

Exercise 05:

```
(kali㉿kali)-[~]
└─$ ccencrypt lab.txt
Enter encryption key:
Enter encryption key: (repeat)

(kali㉿kali)-[~]
└─$ ls
10.125.247.51 Desktop Documents Downloads images.jpeg lab.txt.cpt Music Pictures Public Templates Videos

(kali㉿kali)-[~]
└─$ cdecrypt -d lab.txt.cpt
Enter decryption key:

(kali㉿kali)-[~]
└─$ ls
10.125.247.51 Desktop Documents Downloads images.jpeg lab.txt Music Pictures Public Templates Videos
```

Extracted Information

The file [lab.txt](#) was encrypted using [ccencrypt](#), which requested an encryption key and generated the encrypted file [lab.txt.cpt](#). After that, the file was decrypted using [ccrypt -d lab.txt.cpt](#), and the original file [lab.txt](#) became accessible again.

Analysis and Vulnerability

This exercise shows how file encryption protects data confidentiality by converting a normal file into an unreadable encrypted file that requires the correct key for access. The main risk is not the encryption tool itself, but weak password choice or poor key handling, because anyone who obtains the correct key can decrypt the file.

Exploitation

An attacker who steals the encrypted file alone cannot easily read its content without the decryption key. However, if the attacker guesses, captures, or obtains the key, the protected file can be decrypted and the confidential data exposed.

Protection

Use a strong unique encryption key, never share it insecurely, and store encrypted files separately from the key. This makes file theft less useful to an attacker and improves protection of sensitive data.

Exercise 06:

```
(kali㉿kali)-[~]  
└─$ steghide embed -cf images.jpeg -ef lab.txt  
Enter passphrase:  
Re-Enter passphrase:  
steghide: the cover file is too short to embed the data.
```

```
(kali㉿kali)-[~]  
└─$ ls -lh images.jpeg  
-rw-rw-r-- 1 kali kali 11K Mar  2 10:26 images.jpeg  
  
(kali㉿kali)-[~]  
└─$ ls -lh images\ \ (copy\ 1\).jpeg  
-rw-rw-r-- 1 kali kali 11K Mar  2 10:13 'images (copy 1).jpeg'
```

```
(kali㉿kali)-[~]  
└─$ steghide extract -sf images.jpeg  
Enter passphrase:  
the file "lab.txt" does already exist. overwrite ? (y/n) y  
wrote extracted data to "lab.txt".
```

Extracted Information

[steghide](#) was used with a passphrase to hide [lab.txt](#) inside [images.jpeg](#), and the file sizes were checked with [ls -lh](#). The compared image files have the **same size (11K)**, and the command [steghide extract -sf images.jpeg](#) successfully recovered [lab.txt](#), proving that hidden data can be concealed inside an image and later restored.

Analysis and Vulnerability

This exercise demonstrates data concealment using steganography, where information is hidden inside an image file instead of being stored in a visible form. A key limitation is that the cover image must have enough capacity to store the hidden file. The security risk is that concealed data may pass unnoticed because the image can look normal during casual inspection.

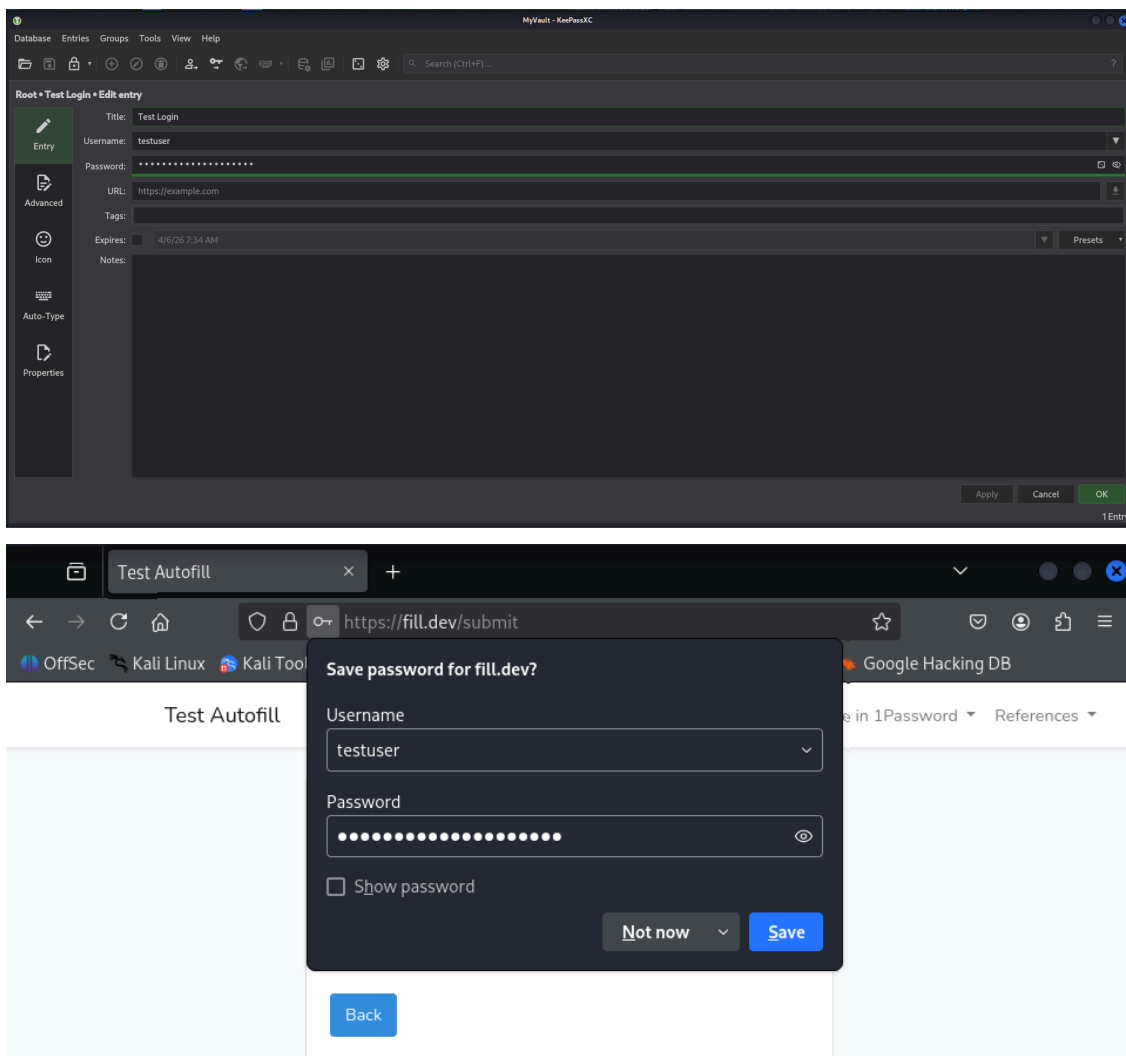
Exploitation

An attacker can hide confidential information, scripts, or small files inside image files and transfer them without attracting attention. If security checks do not inspect media files deeply, the hidden content may remain undetected.

Protection

Use steganalysis or forensic inspection tools on suspicious media files, monitor unauthorized file transfers, and apply strict controls on files exchanged in sensitive environments. Visible file content alone should not be trusted when covert data hiding is possible.

Exercise 07:



Extracted Information

KeePassXC was installed and used to create a password vault named [MyVault](#). A login entry was added for [testuser](#), a strong password was generated and stored in the vault, and the auto-fill feature was tested successfully in the browser.

Analysis and Vulnerability

This exercise demonstrates secure password management using a vault instead of storing passwords manually or reusing weak credentials. The main risk without such a tool is password reuse, weak password selection, and unsafe storage of credentials in plain text or memory.

Exploitation

An attacker can exploit weak or reused passwords to gain unauthorized access to accounts. If credentials are stored insecurely or typed manually into fake pages, they may also be stolen through phishing or local compromise.

Protection

Use a password manager such as KeePassXC, generate unique strong passwords for each account, protect the vault with a strong master password, and use auto-fill carefully only on trusted login pages.

Exercise 08:

```
GNU nano 8.6 /etc/security/pwquality.conf
# Length of substrings from the username to check for in the password
# The check is enabled if the value is greater than 0 and usercheck is enabled.
# usersubstr = 0
#
# Whether the check is enforced by the PAM module and possibly other
# applications.
# The new password is rejected if it fails the check and the value is not 0.
# enforcing = 1
#
# Path to the cracklib dictionaries. Default is to use the cracklib default.
# dictpath =
#
# Prompt user at most N times before returning with error. The default is 1.
# retry = 3
#
# Enforces pwquality checks on the root user password.
# Enabled if the option is present.
# enforce_for_root
#
# Skip testing the password quality for users that are not present in the
# /etc/passwd file.
# Enabled if the option is present.
# local_users_only

minlen = 12
minclass = 3
dcredit = -1
ucrcdit = -1
lcredit = -1
ocredit = -1
maxrepeat = 2
retry = 3
enforcing = 1

(kali@kali)-[~]
$ passwd
Changing password for kali.
Current password:
New password:
BAD PASSWORD: The password contains less than 1 uppercase letters
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
New password:
Retype new password:
passwd: password updated successfully
```

Extracted Information

[pwquality](#) was configured to enforce a strong password policy with a minimum length of [12](#) characters, at least [3](#) character classes, required uppercase, lowercase, digit, and special characters, a repeated character limit of [2](#), and [retry = 3](#). The test with [passwd](#) showed that weak passwords were rejected and a stronger password was accepted successfully.

Analysis and Vulnerability

This exercise demonstrates password policy enforcement on Linux to reduce weak password usage. Without such controls, users may choose simple or predictable passwords that are easier to guess, reuse, or crack.

Exploitation

An attacker can exploit weak passwords through brute-force, dictionary attacks, or credential guessing. If password quality rules are not enforced, the risk of unauthorized access becomes much higher.

Protection

Use [pwquality](#) with strong complexity requirements, enforce password checks through PAM, and require users to choose long and diverse passwords. This reduces the success of password-based attacks.

Exercise 09:

```
(kali㉿kali)-[~]
$ sudo chage -l kali
Last password change           : May 01, 2026
Password expires                : Jul 30, 2026
Password inactive               : never
Account expires                 : never
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
```

Extracted Information

The [chage](#) command was used to display the password aging settings of the user [kali](#). The result shows that the password expires on [Jul 30, 2026](#), the minimum number of days between password changes is [7](#), the maximum number of days is [90](#), and the warning period before expiration is [7](#) days.

Analysis and Vulnerability

This exercise demonstrates password aging enforcement on Linux to limit how long a password can remain valid. Without password expiration rules, users may keep the same password for a long time, which increases the risk if the password is stolen or reused.

Exploitation

An attacker who obtains a valid password can continue using it for a longer period if no expiration policy is enforced. Long password lifetime increases the impact of stolen credentials and makes persistence easier.

Protection

Use [chage](#) to enforce password aging, define a reasonable maximum lifetime such as [90](#) days, set a warning period before expiration, and prevent very frequent password changes with a minimum age. This improves account security and reduces long-term password exposure.

Exercise 10:

```
GNU nano 8.6                                     hibp_check.sh *
#!/usr/bin/env bash

read -rsp "Enter password to check: " PASSWORD
echo

SHA1=$(printf "%s" "$PASSWORD" | shasum | awk '{print toupper($1)}')
PREFIX=${SHA1:0:5}
SUFFIX=${SHA1:5}

echo "SHA1: $SHA1"
echo "Prefix sent to API: $PREFIX"

RESPONSE=$(curl -s -H "user-agent: lab-check" "https://api.pwnedpasswords.com/range/$PREFIX")
MATCH=$(printf "%s\n" "$RESPONSE" | grep "^$SUFFIX:")

if [ -n "$MATCH" ]; then
    COUNT=$(echo "$MATCH" | cut -d: -f2)
    echo "Result: Password found in breaches."
    echo "Pwned count: $COUNT"
else
    echo "Result: Password not found in returned breach data."
fi
```

```
(kali㉿kali)-[~]
$ chmod +x hibp_check.sh

(kali㉿kali)-[~]
$ ./hibp_check.sh
Enter password to check:
SHA1: CBFDAC6008F9CAB4083784CBD1874F76618D2A97
Prefix sent to API: CBFDA
Result: Password found in breaches.
Pwned count: 2254650

(kali㉿kali)-[~]
$ ./hibp_check.sh
Enter password to check:
SHA1: 95A7BC15E5F0E0F78DF832210C53BC22AD4C48D0
Prefix sent to API: 95A7B
Result: Password not found in returned breach data.
```

Extracted Information

A Bash script was created to check test passwords against the Have I Been Pwned password database. The script generated a SHA-1 hash for each password, sent only the hash prefix to the API, and compared the returned suffixes locally. The first tested password was found in breach data with a high occurrence count, while the second tested password was not found in the returned results.

Analysis and Vulnerability

This exercise demonstrates how breached-password checking can identify unsafe passwords without exposing the full password during the query. If a password appears in known breach data, it is weak from a security perspective because it may already be known to attackers and reused in automated attacks.

Exploitation

An attacker can use lists of leaked passwords to perform credential stuffing, password spraying, or account compromise attempts. If users choose passwords that have already appeared in previous breaches, the probability of unauthorized access becomes much higher.

Protection

Avoid passwords that appear in breach databases, use unique strong passwords for every account, and verify password safety before using them. A password manager can help generate and store stronger passwords securely.

Exercise 12:

```
[05/01/26]seed@VM:~/itp-onion-demo$ python3 -m http.server --directory ./public 3000
Serving HTTP on 0.0.0.0 port 3000 (http://0.0.0.0:3000/) ...
```

```
GNU nano 4.8                                     torrc
SocksPort 127.0.0.1:9052
DataDirectory /home/seed/itp-onion-demo/tor_data/
HiddenServiceDir /home/seed/itp-onion-demo/hidden_service/
HiddenServicePort 80 127.0.0.1:3000
```

```
[05/01/26]seed@VM:~$ cat ~/itp-onion-demo/hidden_service/hostname
ndt36lu6kbme3punzlx3ftqpwf3swjsktnbermmkpywo2j7seedwtxqd.onion
[05/01/26]seed@VM:~/onionscan$ ./onionscan scan ndt36lu6kbme3punzlx3ftqpwf3swjsktnbermmkpywo2j7seedwtxqd.onion
Starting embedded Tor daemon...
This may take 1-3 minutes while Tor bootstraps and connects to the network.

Embedded Tor daemon started successfully!
SOCKS proxy: 127.0.0.1:44771

Scanning ndt36lu6kbme3punzlx3ftqpwf3swjsktnbermmkpywo2j7seedwtxqd.onion...
Scan completed in 2m18.232s
```

```
=====
                        ONIONSCAN REPORT
=====

Hidden Service: ndt36lu6kbme3punzlx3ftqpwf3swjsktnbermmkpywo2j7seedwtxqd.onion
Scan Date:      2026-05-01 10:19:02 EDT
Pages Crawled:  0
Status:         Complete
```

SEVERITY SUMMARY

```
-----
CRITICAL: 0
HIGH:     0
MEDIUM:   0
LOW:       0
INFO:     0

TOTAL:    0 findings
```

```
=====
Report generated by OnionScan 2025
https://github.com/nao1215/onionscan
=====
```

```
time=2026-05-01T10:21:21.261-04:00 level=ERROR msg="failed to stop embedded Tor" error="signal: killed"
```

Extracted Information

A local onion service was created and scanned successfully using OnionScan. The target hidden service was: ndt36lu6kbme3punzlx3ftqpwf3swjsktnbermmkpywo2j7seedwtxqd.onion, the scan completed successfully, and the report showed [Pages Crawled: 0](#) with [0 findings](#) across all severity levels. The status of the scan was [Complete](#).

Analysis and Vulnerability

This exercise demonstrates the use of OnionScan to analyze a hidden service and check for exposed metadata or security findings. In this case, the scan completed successfully but did not report any findings, which means no obvious metadata leaks or vulnerabilities were detected from the scan result.

Exploitation

If a hidden service exposes metadata, links, identifiers, or configuration weaknesses, an attacker may use that information to profile the service or reduce its anonymity. In this case, no such findings were reported by the scan.

Protection

Keep the onion service configuration minimal, avoid exposing identifying metadata, and regularly scan hidden services with compatible analysis tools. Using updated tools helps verify that no information is unintentionally leaked.

Exercise 13:

[Suricata IDS/IPS](#) was configured in OPNsense through the Intrusion Detection service and enabled on the **WAN** interface using IPS mode. The downloaded rulesets included the abuse.ch threat intelligence feeds, and a policy was created with the action set to **Drop** in order to block matching malicious traffic. Rule updates were scheduled automatically, and the configuration was verified through the IDS/IPS policy, rules, and alerts pages.

OPNsense® Securing networks made easy root@OPNsense.internal

Lobby Reporting System Interfaces Firewall VPN Services Administration Policy Log File Kea DHCP Monit Network Time OpenDNS Router Advertisements Unbound DNS Power

Settings Download Rules User defined Alerts Schedule

advanced mode

General Settings

Enabled ☒

Capture mode Netmap (IPS)

Promiscuous mode ☐

Interfaces WAN ✖ Clear All ✔ Select All

Detection

Pattern matcher Aho-Corasick, "Ken Steele" variant

Logging

Enable syslog alerts ☐

Enable eve syslog output ☐

Rotate log Weekly

Save logs 4

Apply

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Lobby Reporting System Interfaces Firewall VPN Services Administration Policy Log File Kea DHCP Monit Network Time OpenDNS Router Advertisements Unbound DNS Power

Settings Download Rules User defined Alerts Schedule

Rulesets

Description	Last updated	Enabled	Edit
☒ abuse.ch/Feodo Tracker	2026/05/01 21:14	✔	
☒ abuse.ch/SSL Fingerprint Blacklist	2026/05/01 21:14	✔	
☒ abuse.ch/SSL IP Blacklist	2026/05/01 21:14	✔	
☒ abuse.ch/ThreatFox	2026/05/01 21:14	✔	
☒ abuse.ch/URLhaus	2026/05/01 21:14	✔	
☐ ET open/botcc	not installed	✖	
☐ ET open/botcc.portgrouped	not installed	✖	
☐ ET open/ciarmy	not installed	✖	
☐ ET open/compromised	not installed	✖	
☐ ET open/drop	not installed	✖	

Download & Update Rules

OPNsense® Securing networks made easy root@OPNsense.internal

Lobby Reporting System Interfaces Firewall VPN Services Administration Policy Log File Kea DHCP Monit Network Time OpenDNS Router Advertisements Unbound DNS Power

Service

Rule details

Enabled ☒

Priority 0

Rulesets abuse.ch.feodotracker.rules, abuse.ch.sslblacklist.

Action

Rules

- abuse.ch.feodotracker.rules ☒
- abuse.ch.sslblacklist.rules ☒
- abuse.ch.sslipblacklist.rules ☒
- abuse.ch.threatfox.rules ☒
- abuse.ch.urlhaus.rules ☒
- Nothing selected

first_seen

Nothing selected

New action Drop

Description

Cancel Save

Jobs								
Search 50								
☐ Enabled	Description	Command	Minutes	Hours	Day of the month	Months	Days of the week	Commands
☒	ids rule updates	Update and reload ...	0	0	*	*	*	
« < 1 > » Showing 1 to 1 of 1 entries + □ □								

[Zenarmor](#) was configured in OPNsense as an NGFW layer for application-aware traffic inspection. The protected interface was set to **LAN**, which worked correctly in the lab environment and allowed monitoring of traffic generated by internal client machines. After enabling the engine, traffic was generated from a client VM connected to the same Host-Only network as the OPNsense LAN interface, and the activity was verified through the Zenarmor dashboard and reporting pages.

System: Firmware

Status
Settings
Changelog
Updates
Plugins
Packages

Name	Version	Size	Tier	Repository	Comment	☐ Show community plugins
os-sensei (installed)	2.5	274MiB	2	SunnyValley	Zenarmor is an Enterprise SASE & SSE platform (NGFW, SWG, CASB, ZTNA, SD-WAN)	☐
os-sensei-updater (installed)	2.0	3.75KiB	2	SunnyValley	OPNsense ZENARMOR Plugin Updater	☐
os-sunnyvalley (installed)	1.5_2	2.43KiB	2	OPNsense	Vendor Repository for Zenarmor - Enterprise SASE & SSE platform (NGFW, SWG, CASB, ZTNA, SD-WAN)	☐

OPNsense
Securing networks made easy

Zenarmor Subscription: Free
root@OPNsense.internal

- Lobby
- Reporting
- System
- Interfaces
- Firewall
- VPN
- Services
- Zenarmor
 - Dashboard
 - Reports
 - Live Sessions
 - Policies
 - Settings
 - Notifications
- Power
- Help

Want more power?

Top Threats

No detected threats today

Top Hosts

100%

Top Apps

100%

Download

83.3 Kb

23:51:20 23:51:30 23:51:40 23:51:50 23:52:00 23:52:10

Upload

83.3 Kb

23:51:20 23:51:30 23:51:40 23:51:50 23:52:00 23:52:10

Engine

Status: Running

Version: 2.5 - Apr 22, 2026 12:27

Application DB: 2.0.26042014 - Apr 22, 20...

Start on boot: ☒

Reporting Database

Status: Running

Type: SQLite

Version: 3.50.4

Cloud Agent

Status: Not Installed

Version:

Start on boot: ☐

Please choose interfaces to protect
☒ Show WAN interfaces
Refresh Interfaces

☐ em0 (WAN) wan

☒ em1 (LAN) lan

Zenarmor Subscription: Free
root@OPNsense.internal

Lobby
Reporting
System
Interfaces
Firewall
VPN
Services
Zenarmor
Dashboard
Reports
Live Sessions
Policies
Settings
Notifications
Power
Help

Connections
Threats
Blocks
Web
DNS
TLS

Layout
Filter
Last 6 hours
Pause
100 Rows

Download

Start	End	Device	Device category	Src hostname	Src port	Dest hostname	Dest port
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	64127	192.168.1.1	443
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	53509	192.168.1.1	443
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	65219	192.168.1.1	443
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	51260	192.168.1.1	443
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	63706	192.168.1.1	443
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	63705	192.168.1.1	443
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	50397	192.168.1.1	443
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	60368	192.168.1.1	443
May 1, 2026 23:52	May 1, 2026 23:52	Device (0a0027000007)	Others	192.168.1.2	58946	192.168.1.1	443

Want more power?

[ClamAV](#) was installed in OPNsense as the antivirus engine for scanning malicious content and suspicious files. The **clamd** and **freshclam** services were enabled, and the proxy and ICAP antivirus integration were configured to support scanning of web traffic and downloaded files through the firewall.

Status	Settings	Changelog	Updates	Plugins	Packages
NameVersionSizeTierRepositoryCommentShow community plugins					
os-c-icap (installed)1.949.8KiB3OPNsensec-icap connects the web proxy with a virus scanner					
os-clamav (installed)1.8.147.8KiB3OPNsenseAntivirus engine for detecting malicious threats					
os-sensei (installed)2.5274MiB2SunnyValleyZenarmor is an Enterprise SASE & SSE platform (NGFW, SWG, CASB, ZTNA, SD-WAN)					
os-sensei-updater (installed)2.03.75KiB2SunnyValleyOPNsense ZENARMOR Plugin Updater					
os-squid (installed)1.4296KiB3OPNsenseSquid is a caching proxy for the web					

**OPNsense**
Securing networks made easy

root@OPNsense.internal

Interfaces

Firewall

VPN

Services

C-ICAP

Captive Portal

ClamAV

Configuration

Log File

DHCRelay

Dnsmasq DNS & DHCP

Intrusion Detection

Kea DHCP

Monit

Network Time

OpenDNS

Router Advertisements

Services: ClamAV: Configuration

General

Signatures

Versions

full help

1 Enable clamd service

2 Enable freshclam service

3 Enable TCP port

4 Maximum number of threads running

5 Maximum number of queued items

6 Idle timeout

7 Max directory recursion

8 Follow directory symlinks

9 Follow regular file symlinks

OPNsense (c) 2014-2026 Deciso B.V.

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Firewall

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DHCRelay

Dnsmasq DNS & DHCP

Intrusion Detection

Kea DHCP

Monit

Network Time

OpenDNS

Router Advertisements

Unbound DNS

Services: C-ICAP: Configuration

General

Antivirus

full help

1 Enable c-icap service

2 Timeout

3 Max keepalive requests

4 Max keepalive timeout

5 Start servers

6 Max servers

7 Min spare threads

8 Max spare threads

OPNsense (c) 2014-2026 Deciso B.V.

Services: Squid Web Proxy: Administration

General Proxy Settings

Forward Proxy

Proxy Auto-Config

Remote Access Control Lists

Support

advanced mode

full help

1 Enable proxy

2 User error pages

Squid

Apply

Services: Squid Web Proxy: Administration

General Proxy Settings

Forward Proxy

Proxy Auto-Config

Remote Access Control Lists

Support

advanced mode

full help

Proxy interfaces

LAN

Clear All

Select All

Proxy port

3128

Enable Transparent HTTP proxy

☐

Enable SSL inspection

☐

Log SNI information only

☐

SSL Proxy port

3129

CA to use

Nothing selected

SSL no bump sites

Services: Squid Web Proxy: Administration

General Proxy Settings

Forward Proxy

Proxy Auto-Config

Remote Access Control Lists

Support

advanced mode

full help

Enable ICAP

☒

Request Modify URL

icap://[::1]:1344/avscan

Response Modify URL

icap://[::1]:1344/avscan

Exclusion List

Clear All

Copy

Paste

Text

Apply

Extracted Information

OPNsense was used as a firewall platform and extended with three security layers: **Suricata** for IDS/IPS, **Zenarmor** for next-generation traffic inspection, and **ClamAV** for antivirus protection. Suricata was configured on the WAN side for network threat detection and blocking, Zenarmor was configured on the LAN side for application-aware inspection of client traffic, and ClamAV was enabled to add antivirus support for web traffic and transferred files.

Analysis and Vulnerability

This exercise shows how OPNsense can be transformed from a basic firewall into a layered security gateway. A normal firewall can filter ports and addresses, but without IDS/IPS, NGFW inspection, and antivirus scanning, malicious traffic, suspicious applications, and harmful files may still pass through more easily. Using these three services together improves visibility and strengthens network defense.

Exploitation

An attacker can use reconnaissance traffic, exploit attempts, malicious URLs, command-and-control communication, or malware delivery to compromise systems if only basic filtering is enabled. Without deeper inspection, these activities may look like normal traffic and remain undetected.

Protection

Use Suricata to detect and block suspicious network activity, Zenarmor to monitor and control application-level traffic, and ClamAV to inspect malicious content and downloaded files. Keeping rules, threat feeds, and antivirus signatures updated improves detection and provides stronger protection against attacks and malicious traffic.

Exercise 18:

```
vboxuser@Ubu:~$ sudo systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-05-03 15:31:14 UTC; 44s ago
```

```
vboxuser@Ubu:~$ sudo systemctl status kibana
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-05-03 15:29:46 UTC; 2min 25s ago
```

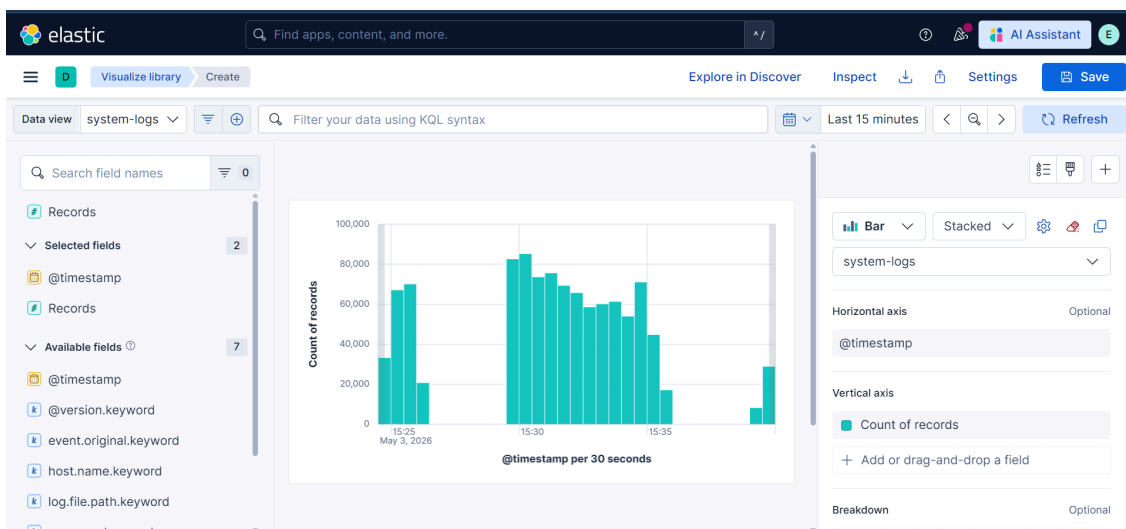
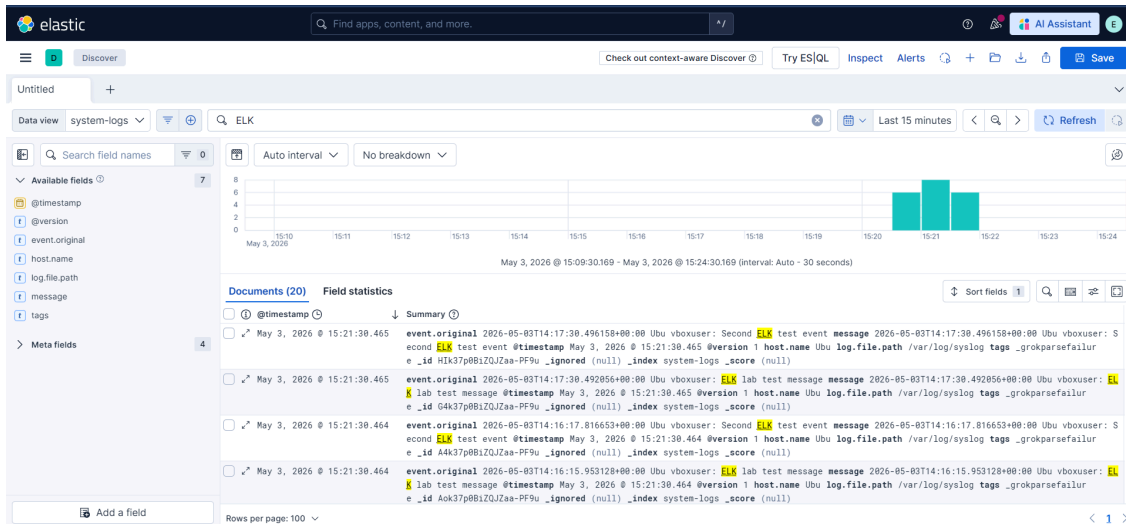
```
vboxuser@Ubu:~$ sudo systemctl status logstash
● logstash.service - logstash
   Loaded: loaded (/usr/lib/systemd/system/logstash.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-05-03 15:35:13 UTC; 1min 49s ago
```

```
GNU nano 7.2 /etc/logstash/conf.d/system-logs.conf
input {
  file {
    path => "/var/log/syslog"
    start_position => "beginning"
    sincedb_path => "/dev/null"
  }
}

filter {
  grok {
    match => { "message" => "%{SYSLOGTIMESTAMP:syslog_timestamp} %{HOSTNAME:host} %{DATA:program}(?:\\[[%{POSINT:pid}\\]]?: %{GREEDYDATA:log_message})" }
  }
}

output {
  elasticsearch {
    hosts => ["https://localhost:9200"]
    index => "system-logs"
    user => "elastic"
    password => "Zj_osEeUYIqY6gp*0Ao7"
    ssl_verification_mode => "none"
  }
  stdout { codec => rubydebug }
}
```

```
vboxuser@Ubu:~$ logger "ELK lab test message"
vboxuser@Ubu:~$ logger "Second ELK test event"
vboxuser@Ubu:~$ logger "User generated a test log for Kibana"
```



Extracted Information

The ELK Stack was installed successfully, including **Elasticsearch**, **Logstash**, and **Kibana**. Test log entries were generated from the system using the [logger](#) command, collected through Logstash, indexed in Elasticsearch, and displayed in Kibana through the Discover interface.

Analysis and Vulnerability

This exercise demonstrates centralized log collection and visualization using the ELK Stack. Centralized logging improves visibility into system activity and helps identify

suspicious behavior, failed actions, or operational issues that may not be noticed when logs remain distributed across individual systems.

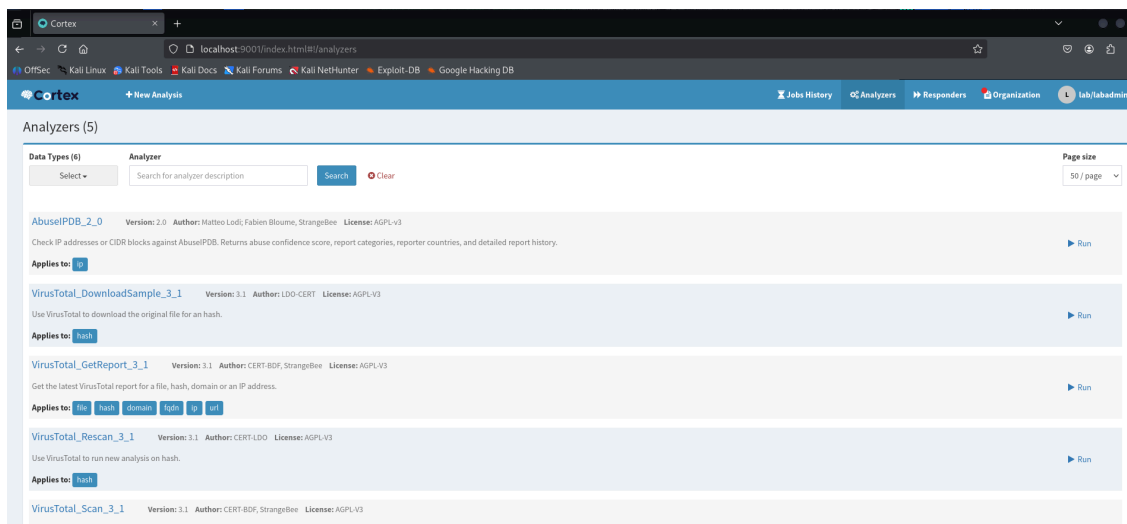
Exploitation

An attacker may benefit when logs are not collected centrally, because malicious actions can remain hidden across multiple systems and be harder to correlate. Without proper log visibility, detection and investigation become more difficult.

Protection

Use centralized logging with Elasticsearch, Logstash, and Kibana to collect, review, and analyze system events in one place. This improves monitoring, supports incident detection, and helps administrators investigate suspicious activity more effectively.

Exercise 19:



Extracted Information

Cortex was installed and configured as the local analysis platform for security observables such as IP addresses and file hashes. The Cortex dashboard confirmed that the service was running, and the analyzers page showed the available analysis modules. A Shuffle workflow was also created to represent the SOAR automation process for sending observables to an analysis action.

Analysis and Vulnerability

This exercise demonstrates the concept of SOAR automation, where observables such as suspicious IP addresses or hashes can be passed automatically to an analysis platform instead of being checked manually. In this implementation, Cortex was prepared as the analyzer backend, and Shuffle was used to model the automation workflow.

Exploitation

Without automated observable analysis, attackers may benefit from slow manual investigation. Suspicious IP addresses, malware hashes, or indicators of compromise may not be checked quickly enough, allowing malicious activity to continue longer.

Protection

Use SOAR workflows to automate observable enrichment and analysis. Connecting Shuffle with Cortex allows IP addresses and hashes to be analyzed faster, improves incident response speed, and reduces repetitive manual investigation tasks.

Exercise 25:

NXLog Configuration: Because Windows does not provide native Syslog forwarding by default, **NXLog** was used to collect Windows Security and System logs and forward them to the Kali VM over Syslog.

```
PS C:\Windows\System32> Get-Service nxlog

Status Name          DisplayName
-----
Running nxlog         nxlog
```

```
define ROOT C:\Program Files\nxlog

Moduledir %ROOT%\modules
Cachedir %ROOT%\data
Pidfile %ROOT%\data\nxlog.pid
SpoolDir %ROOT%\data
Logfile %ROOT%\data\nxlog.log

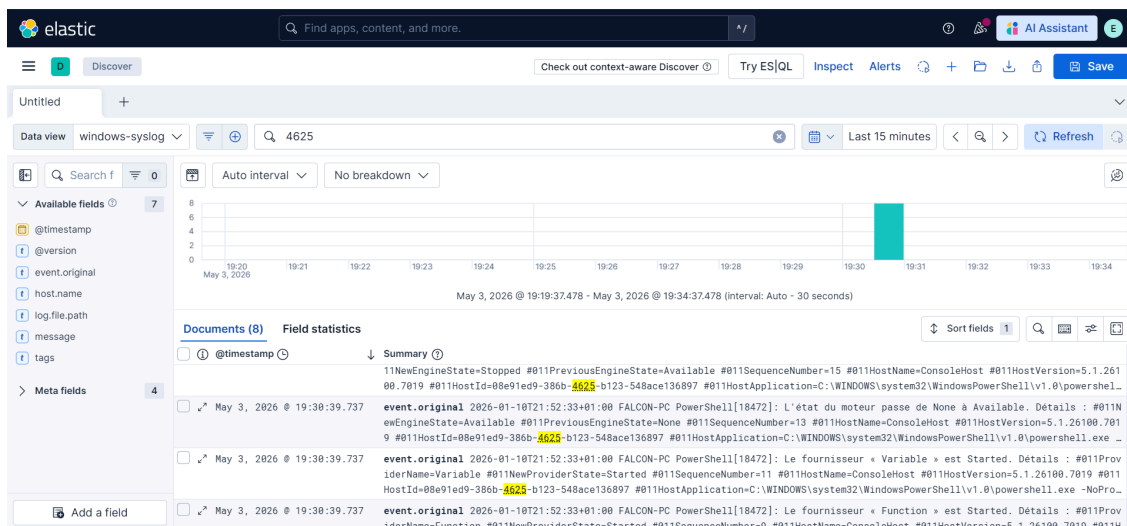
<Extension syslog>
  Module xm_syslog
</Extension>

<Input eventlog>
  Module im_msvistalog
</Input>

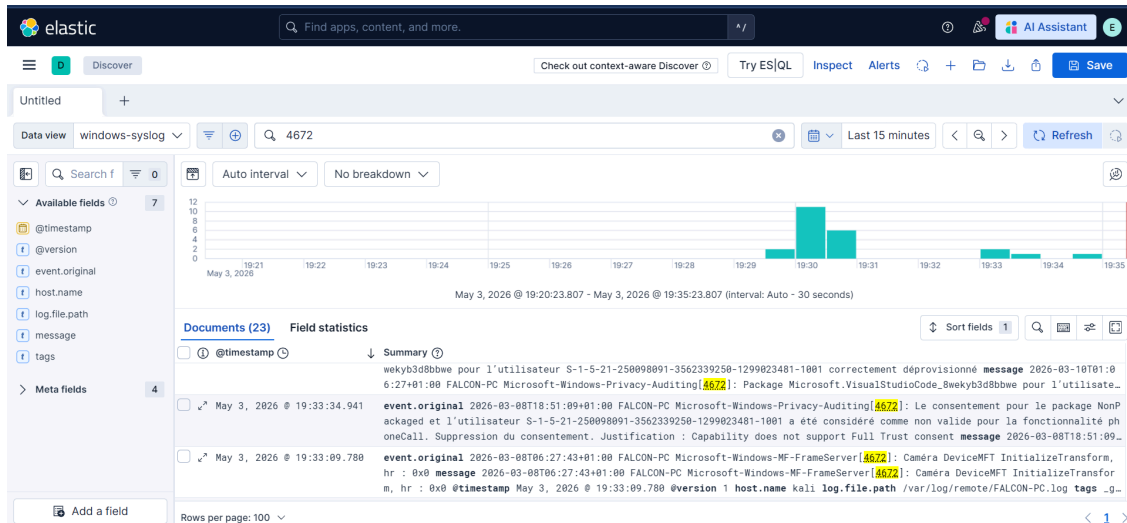
<Output out>
  Module om_udp
  Host 192.168.238.101
  Port 514
  Exec to_syslog_bsd();
</Output>

<Route r>
  Path
</Route>
```

Failed Login Events: In Kibana Discover, event **4625** was identified in the forwarded Windows logs. This event corresponds to failed login attempts and was used as an anomaly indicator for authentication-related suspicious activity.



Privilege-Related Events: In Kibana Discover, event **4672** was identified in the forwarded Windows logs. This event indicates that special privileges were assigned to a new logon session and was used as an anomaly indicator for privileged activity.



Extracted Information

In this exercise, Windows event logs were forwarded from the host machine to the Kali VM using **NXLog** over Syslog. On Kali, **rsyslog** received the remote logs, **Logstash** processed them, **Elasticsearch** indexed them, and **Kibana** was used to visualize and analyze them through the [windows-syslog*](#) data view. During the analysis, event **4625** was identified as failed login activity and event **4672** was identified as a privilege-related event.

Analysis and Vulnerability

This exercise demonstrates a complete log analysis workflow using **Syslog + ELK Stack**. Because Windows does not natively send logs using Syslog, an additional forwarding tool was required to integrate Windows logs into the ELK pipeline. By forwarding the logs with NXLog to Kali and visualizing them in Kibana, it became possible to detect suspicious authentication and privilege-related activity centrally.

Exploitation

An attacker may generate repeated failed logins during password guessing or brute-force attempts, which appear as event **4625**. If elevated privileges are obtained or used, privilege-related events such as **4672** may appear. Centralized collection and analysis make these attack indicators easier to detect and investigate.

Protection

Use **NXLog** or a similar forwarding agent to export Windows logs, receive them through **Syslog**, and analyze them with **Elasticsearch**, **Logstash**, and **Kibana**. Monitoring important Windows event IDs such as **4625** and **4672** helps detect failed authentication attempts, privileged activity, and possible unauthorized access more effectively.

Conclusion

This lab provided practical experience with several important cybersecurity tools and techniques used for protection, monitoring, and incident response. The exercises covered network scanning, vulnerability identification, encryption, steganography, password security, breach checking, onion service analysis, firewall configuration, IDS/IPS deployment, antivirus integration, log collection, anomaly detection, and ticketing.

The lab showed that effective cybersecurity depends on multiple layers of defense. Nmap helped identify exposed services and understand the attack surface, while encryption and steganography demonstrated methods for protecting or hiding data. Password-related exercises highlighted the importance of strong credentials, password policies, password aging, and checking for breached passwords. OPNsense, Suricata, Zenarmor, and ClamAV demonstrated how a firewall can be strengthened with detection, prevention, application inspection, and antivirus capabilities.

The ELK Stack exercises showed the importance of centralized logging and visualization for detecting suspicious activity, such as failed logins or privilege-related events. TheHive demonstrated how incident cases can be documented, assigned, and tracked to improve accountability and traceability.

Overall, this lab connected theoretical cybersecurity concepts with real implementation and showed how different tools work together to reduce risk, detect threats, and support organized incident response.